Script-Controlled ACL – Restrict Record Access Based on Field Value

Description :

This lab demonstrates how to implement a **script-controlled Access Control List (ACL)** to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the **EEE branch** are allowed to view records where the **Branch field is set to EEE**, while administrators retain full access.

Timer :

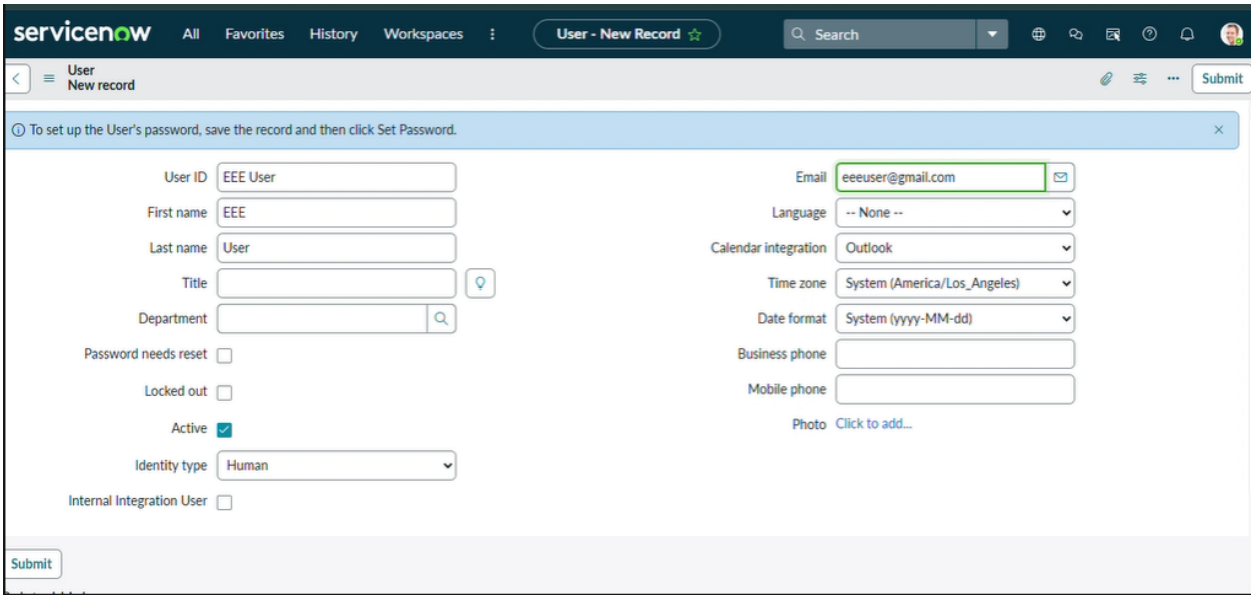
2 hrs

Project Overview:

Milestone-1 Creation of Users and Roles

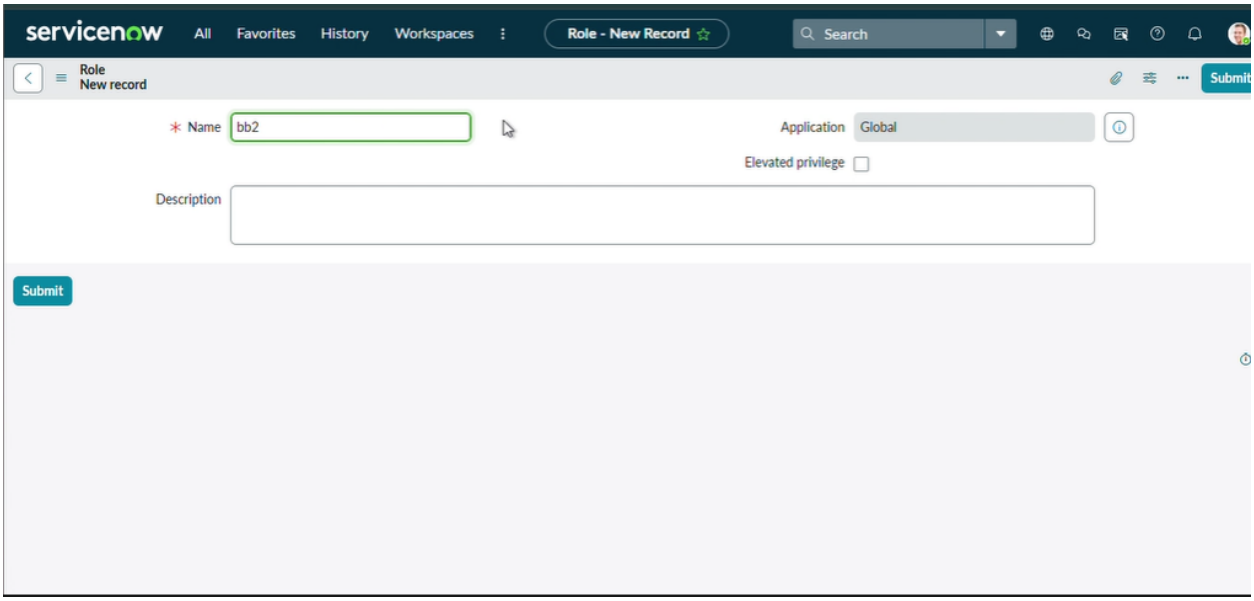
User Creation:

1. **Log in** to the ServiceNow instance with **admin access**.
2. **Create a test user:**
    - Navigate to **User Administration → Users**
    - Click **New**
    - **User ID**: EEE User
    - **First name**: EEE
    - **Last name**: User
    - **Email**: eeeuser@gmail.com
    - Save the record



Role Creation

- Navigate to **User Administration → Roles**
- Click **New**
- **Name**: Enter role name as bb1
- **Description**: Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User



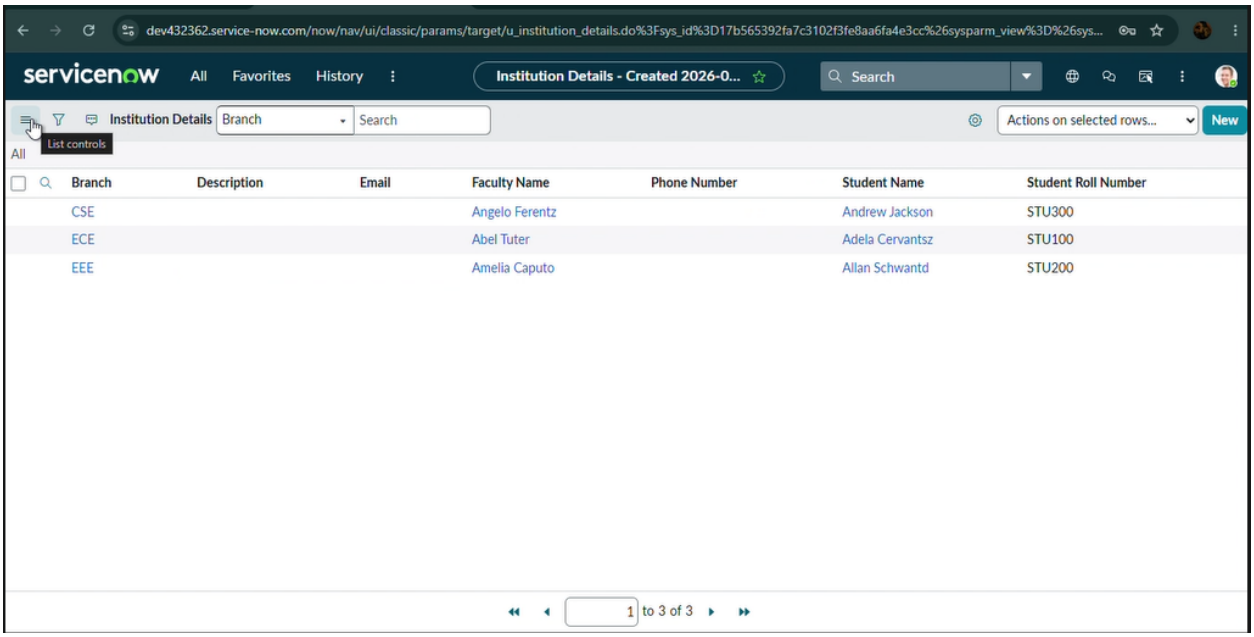
Tables Creation

1. In the **Application Navigator**, search for **System Definition → Tables**.
2. Click **New** and create a custom table with the following details:
    - **Label:** Institution Details
    - **Name:** u_institution_details
    - **Extends for :** false



3. Click **Submit**.
4. Add the following fields to the **u_institution_details** table:
    - **Student Roll Number**– Auto Number
    - **Student Name** – Reference- User
    - **Faculty Name** – Reference- User
    - **Branch** – Choice (ECE, EEE, CSE)
    - **Email**– String
    - **Phone Number**– String
    - **Description**– Multi String
5. Save the Table

6. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).



Creation of Access Control Lists(Read ACL)

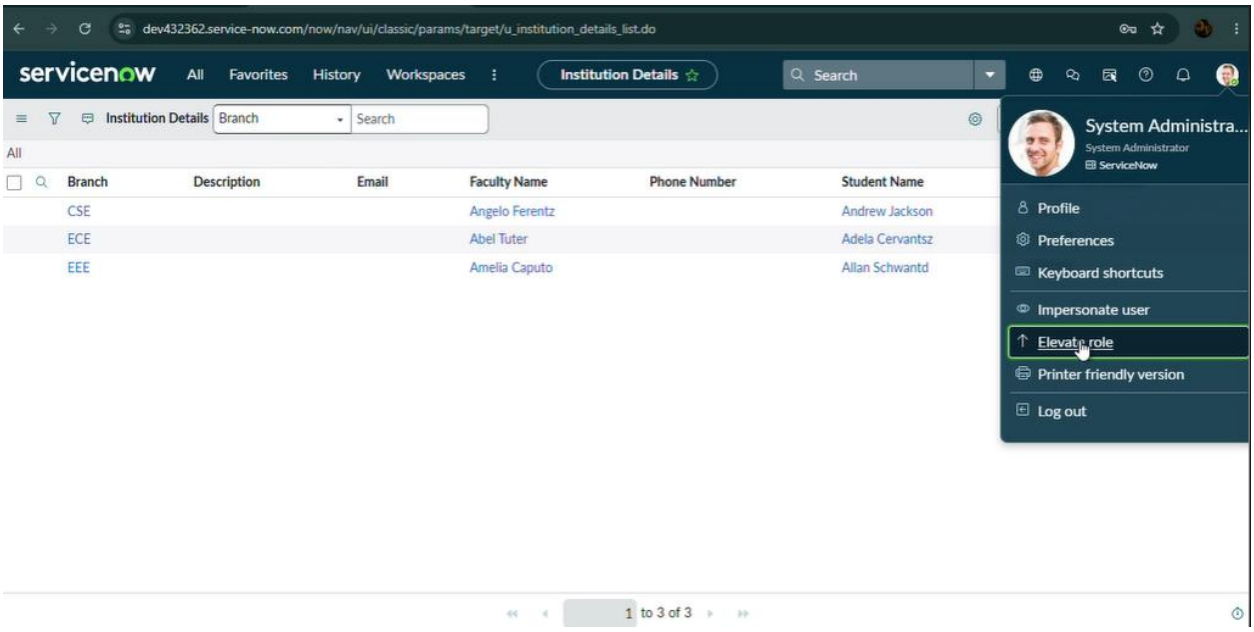
1. In the **Application Navigator**, search for **System Security → Access Control (ACL)**.
2. Elevate "Security_admin" role
3. Click **New** to create a new ACL.
4. Enter the following details:
    - **Type:** record
    - **Operation:** read
    - **Name:** u_institution_details
    - **Active:** true
    - **Advanced:** true
5. In the **Requires role** related list, add:
    - bb1 (custom role)
6. Add Data condition as **Branch is EEE**
7. In the **Script** field, add the following script:

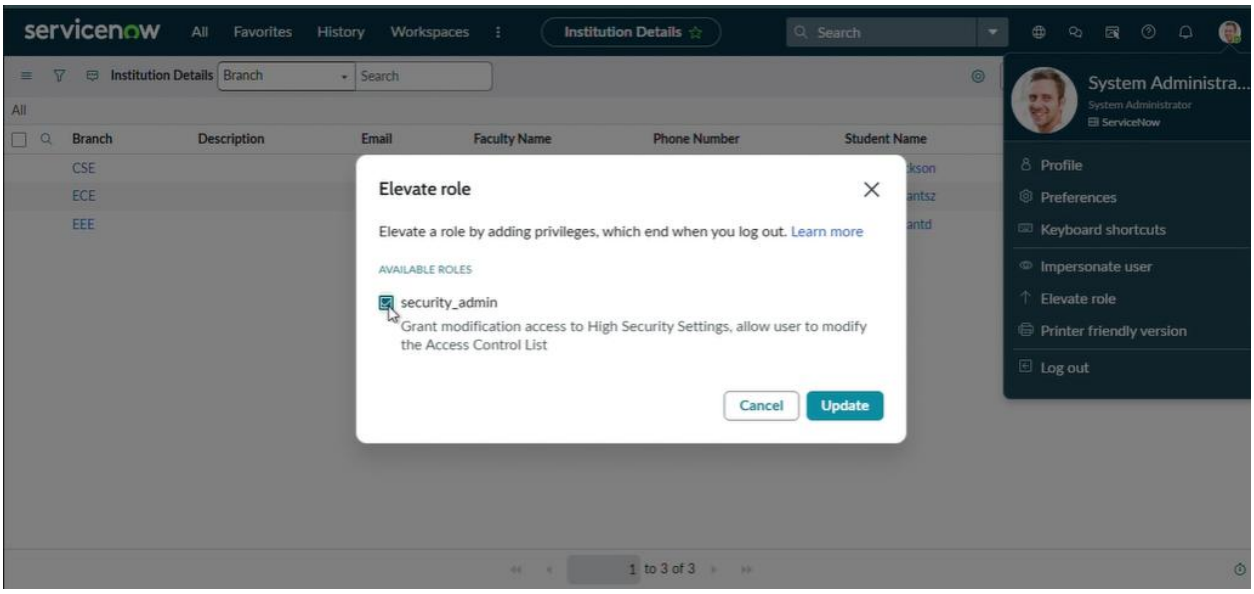
```
(function () {

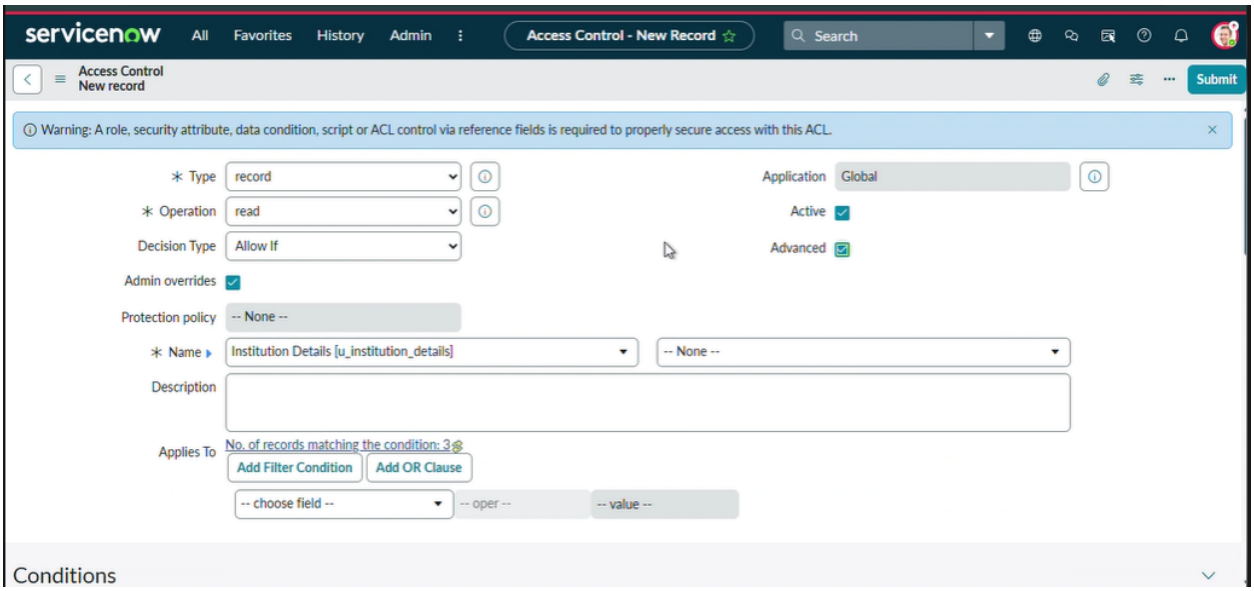
// Allow admin users full access
if (gs.hasRole('admin')) {
   return true;
}
// Allow only EEE branch users to see EEE records
if (gs.hasRole('bb1')){
   return true;
```

```
    }
    // Deny access for all others
    return false;
})();
```

8. Click **Submit** to save the ACL.



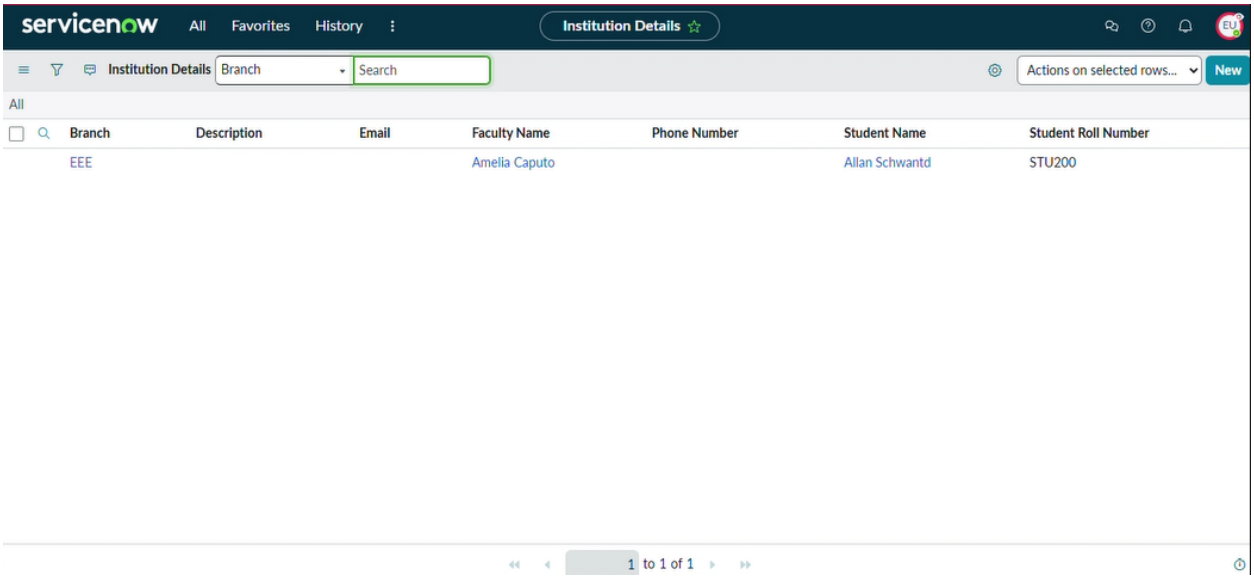




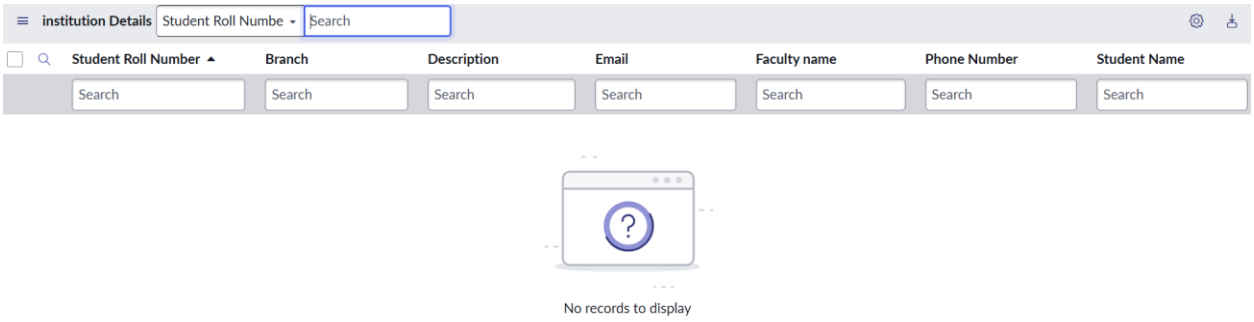
9. Impersonate **EEE users** and open the Student Records list.
10. Copy the table name u_institution_details.list/LIST and paste in the All menu.
11. Impersonate **Admin** and open the Student Records list.

Verify :

- User with **bb1 role** can view **only EEE branch records**

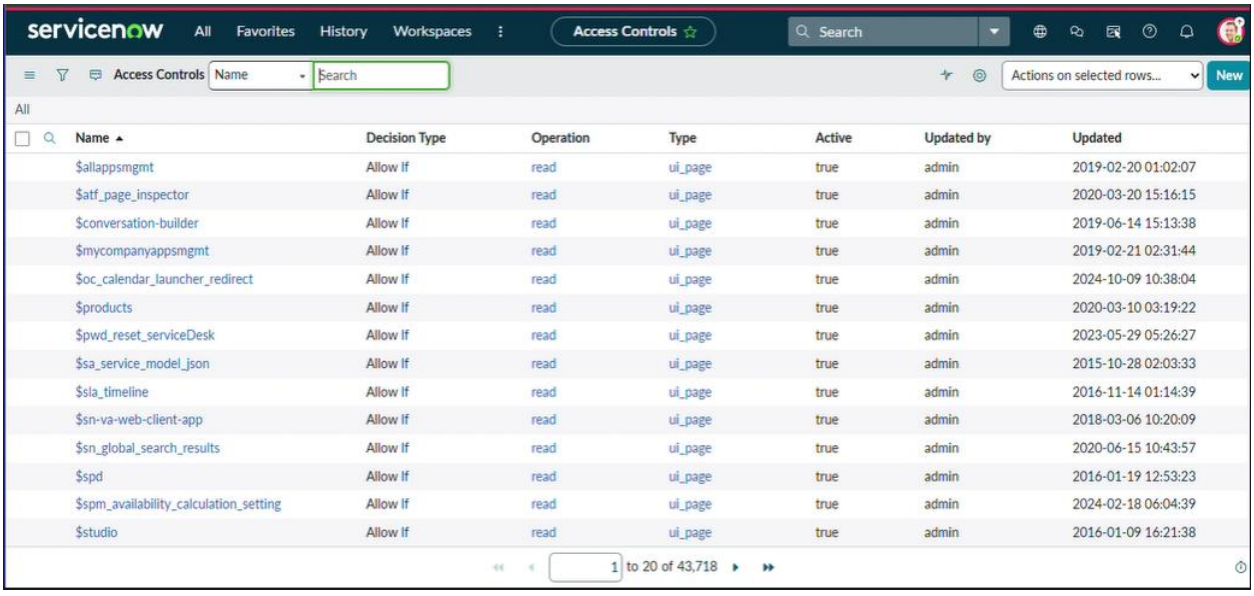


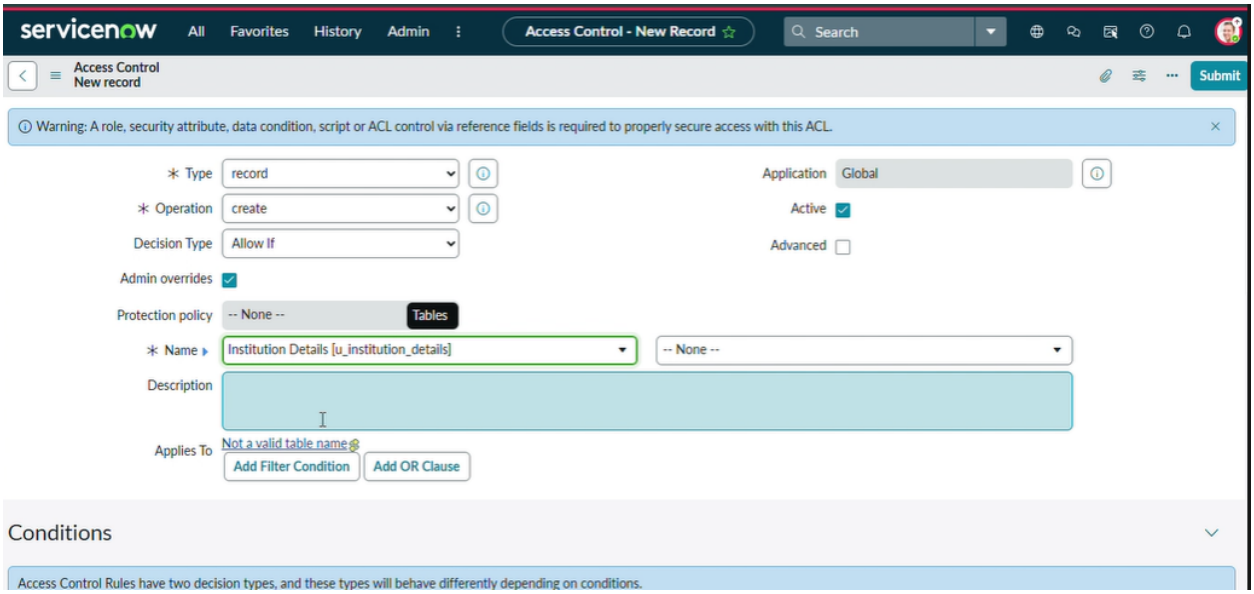
- User without the role cannot view any records

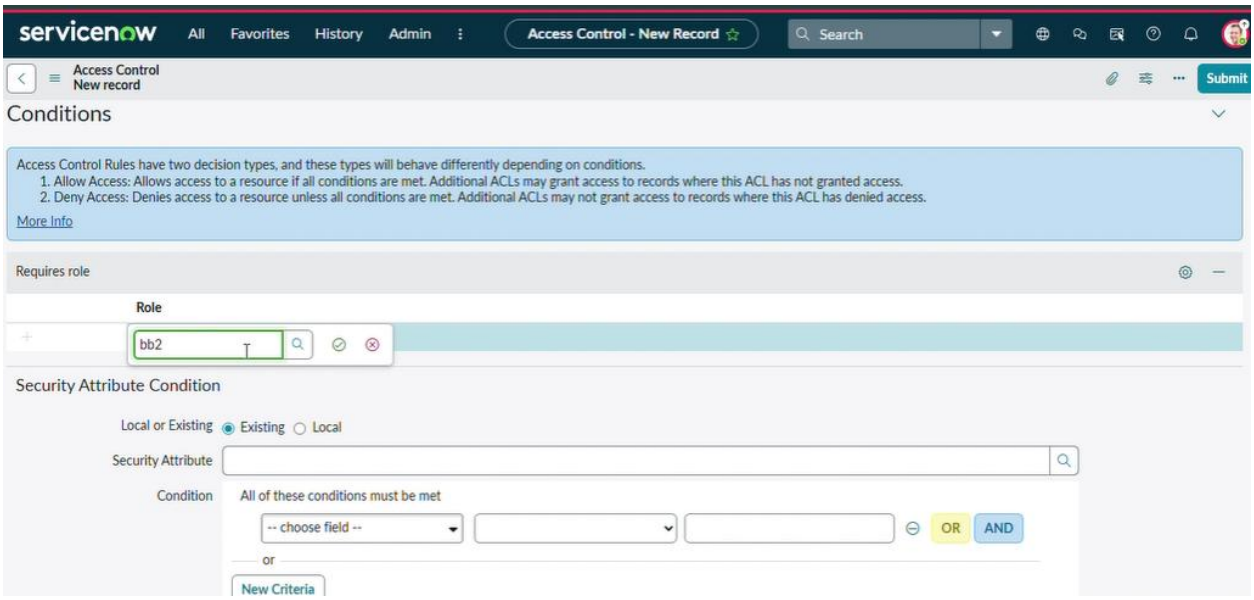


Creation of Access Control Lists(Create ACL)

1. Click **New** to create a new ACL(Create)
2. Enter the following details:
    - **Type:** record
    - **Operation:** create
    - **Name:** u_institution_details
    - **Active:** true
3. In the **Requires role** related list, add:
    - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record

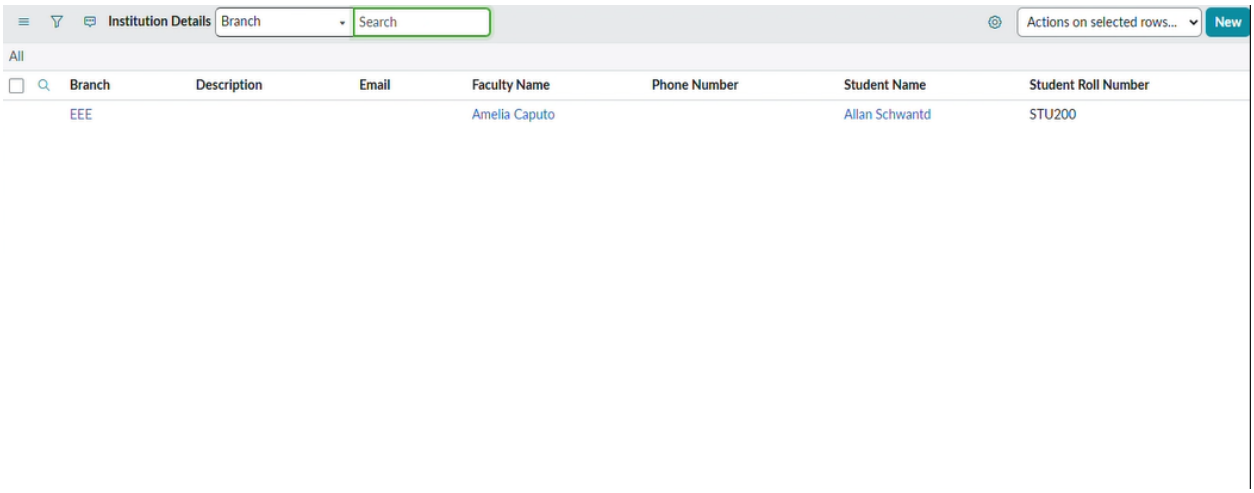






Verify :

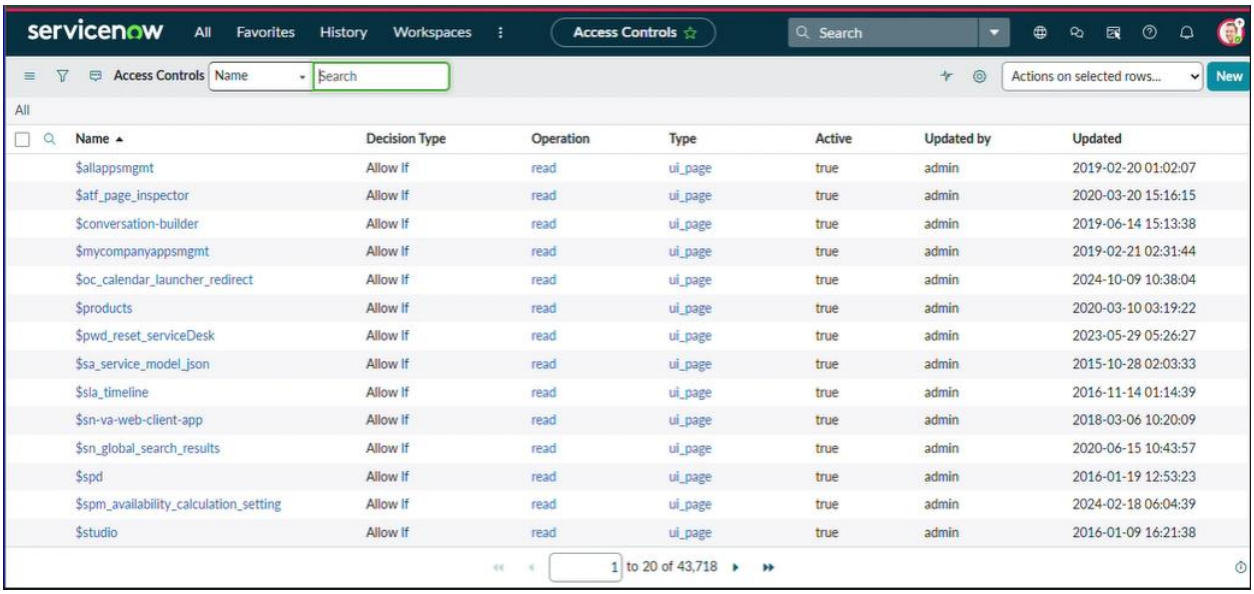
- User with **bb1 &bb2 roles** can view **only EEE branch records b New** Button

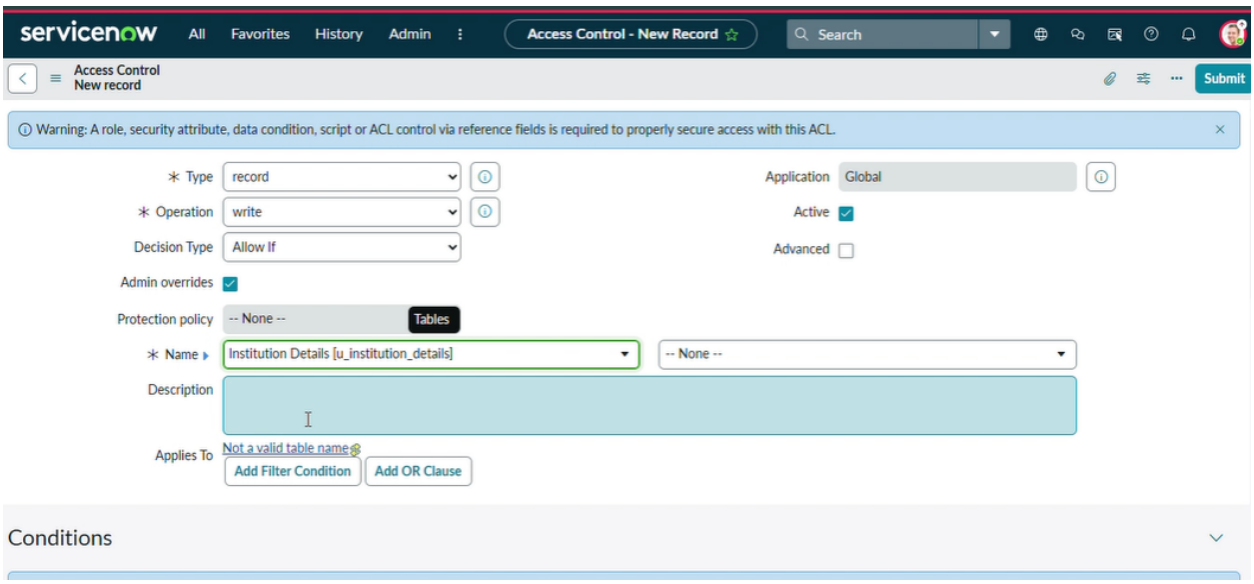


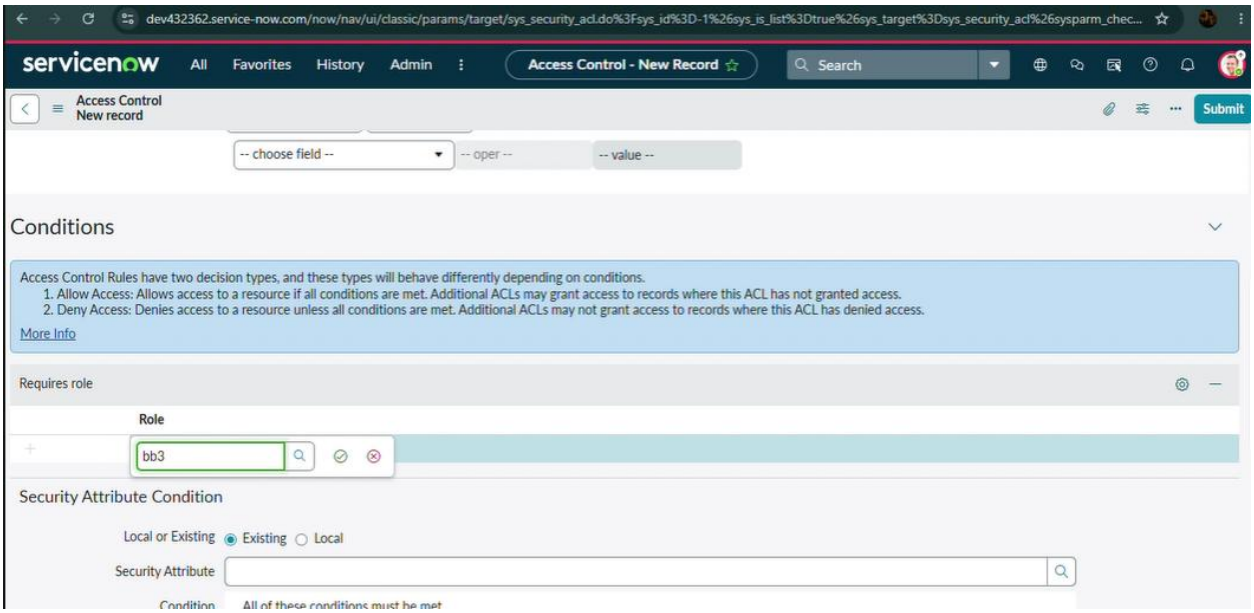
Creation of Access Control Lists(Write ACL)

12. Click New to create a new ACL(Write)
13. Enter the following details:
    - Type: record
    - Operation: Write
    - Name: u_institution_details
    - Active: true
14. In the Requires role related list, add:
    - bb3 (custom role)
15. No need to add any Data condition
16. Save the ACL Record



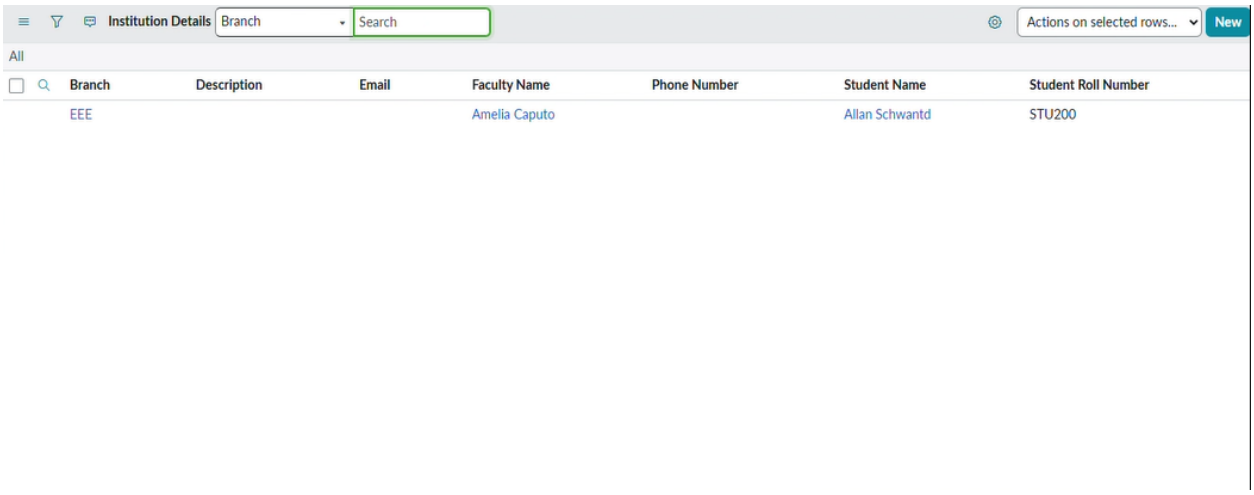

servicenow
All
Favorites
History
Workspaces
Access Controls
Search
Access Controls
Name
Search
Actions on selected rows...
New
All
Name
Decision Type
Operation
Type
Active
Updated by
Updated
$allappsmgmt Allow If read ui_page true admin 2019-02-20 01:02:07
$atf_page_inspector Allow If read ui_page true admin 2020-03-20 15:16:15
$conversation-builder Allow If read ui_page true admin 2019-06-14 15:13:38
$mycompanyappsmgmt Allow If read ui_page true admin 2019-02-21 02:31:44
$oc_calendar_launcher_redirect Allow If read ui_page true admin 2024-10-09 10:38:04
$products Allow If read ui_page true admin 2020-03-10 03:19:22
$pwd_reset_serviceDesk Allow If read ui_page true admin 2023-05-29 05:26:27
$sa_service_model_json Allow If read ui_page true admin 2015-10-28 02:03:33
$sla_timeline Allow If read ui_page true admin 2016-11-14 01:14:39
$sn-va-web-client-app Allow If read ui_page true admin 2018-03-06 10:20:09
$sn_global_search_results Allow If read ui_page true admin 2020-06-15 10:43:57
$spd Allow If read ui_page true admin 2016-01-19 12:53:23
$spm_availability_calculation_setting Allow If read ui_page true admin 2024-02-18 06:04:39
$studio Allow If read ui_page true admin 2016-01-09 16:21:38
1 to 20 of 43,718


servicenow
All
Favorites
History
Admin
Access Control - New Record
Search
Access Control
New record
Submit
Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.
Type
record
Application
Global
Operation
write
Active
Decision Type
Allow If
Advanced
Admin overrides
Protection policy
-- None --
Tables
Name
Institution Details [u_institution_details]
-- None --
Description
Applies To
Not a valid table name
Add Filter Condition
Add OR Clause
Conditions



Verify :

- User with **bb1 &bb2 &bb3 roles** can view and edit the **EEE branch records along with New** Button



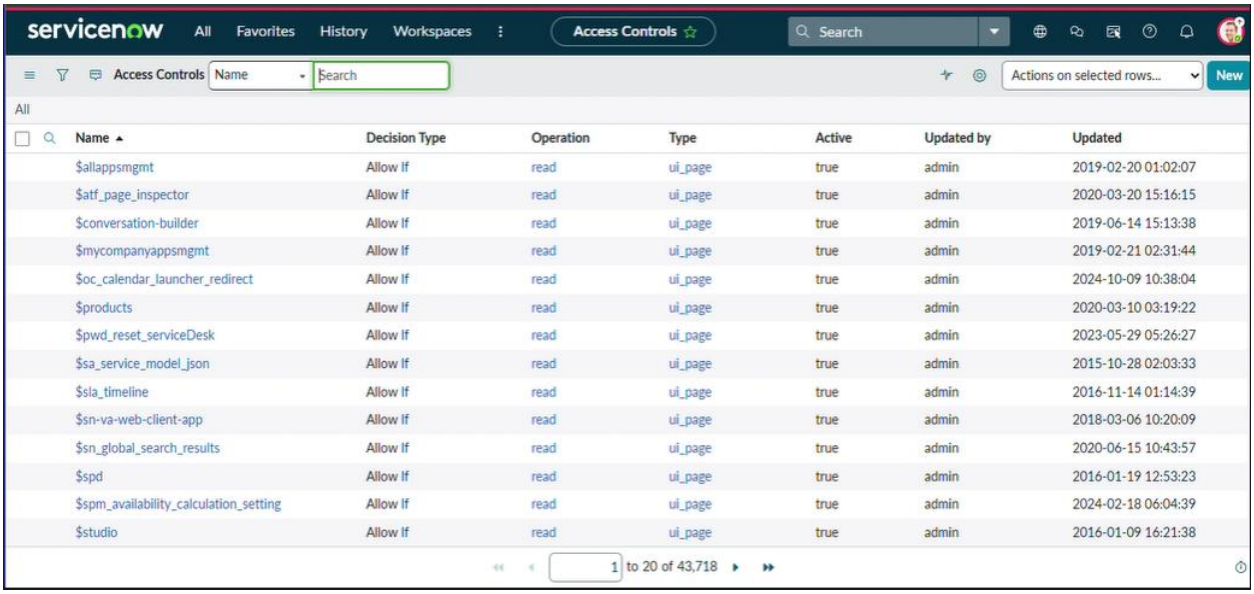
Creation of Access Control Lists(Delete ACL)

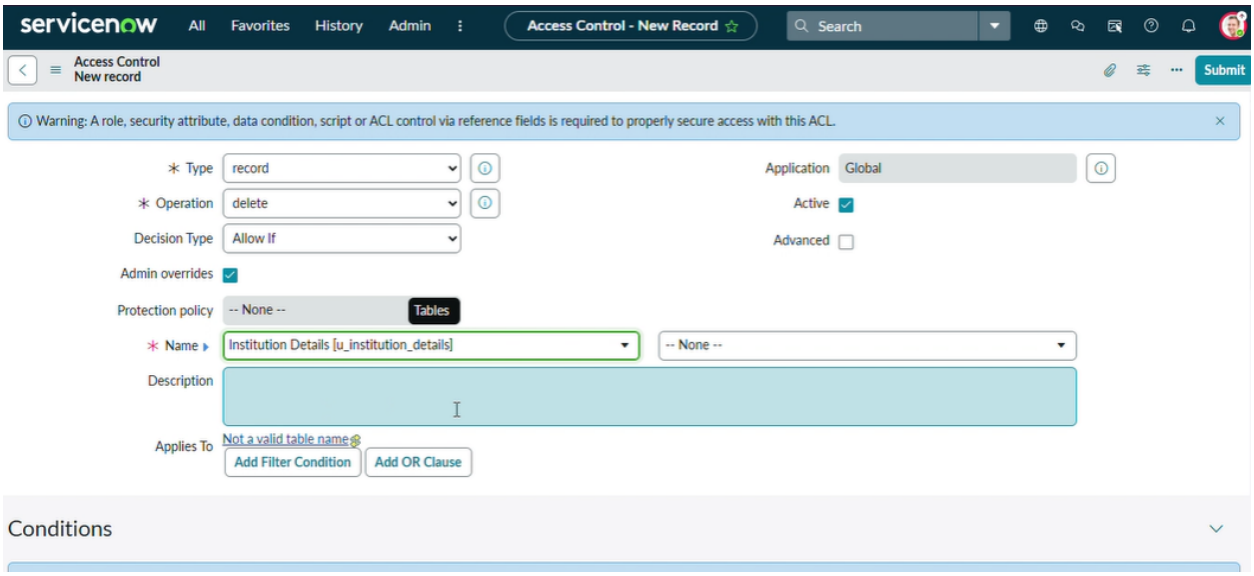
17. Click **New** to create a new ACL(Delete)
18. Enter the following details:
    - **Type:** record

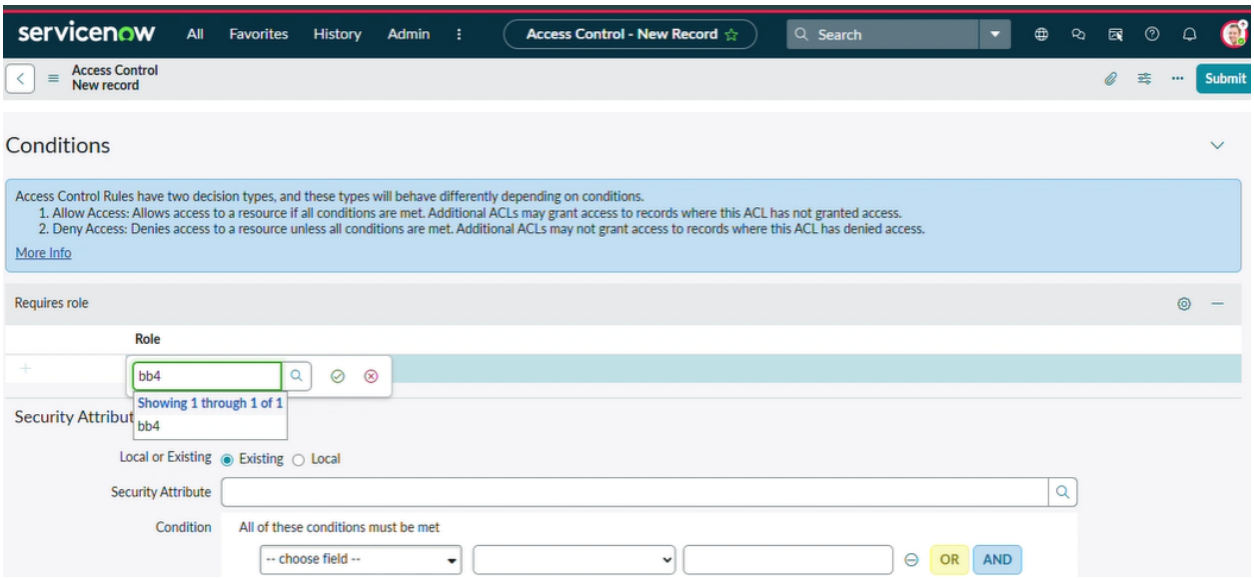
- **Operation:** Delete
- **Name:** u_institution_details
- **Active:** true

19. In the **Requires role** related list, add:
    - bb4 (custom role)
20. No need to add any Data condition
21. Save the ACL Record



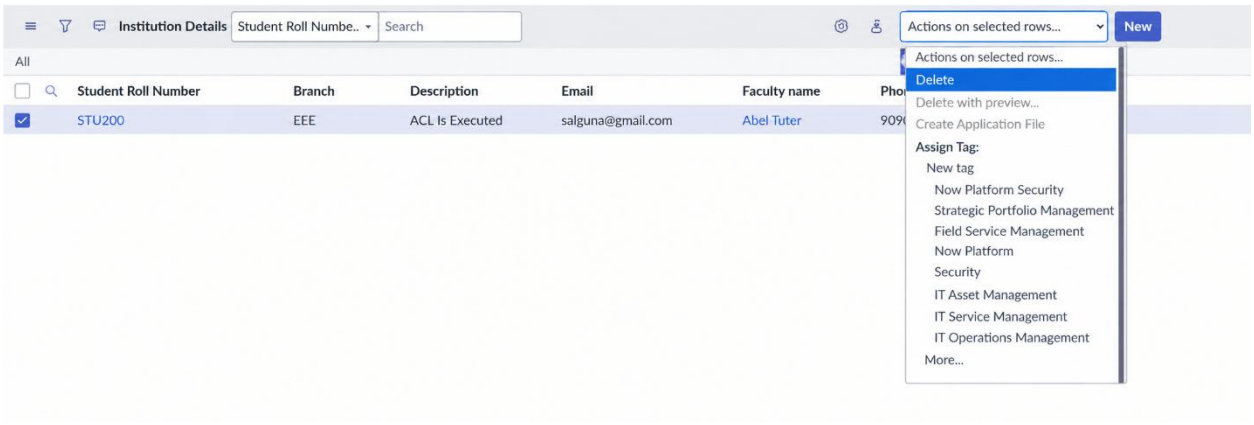






Verify :

- Users with **bb1 &bb2 &bb3&bb4 roles** can see, create and edit along with delete the **EEE branch records.**



Outcome :

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.